

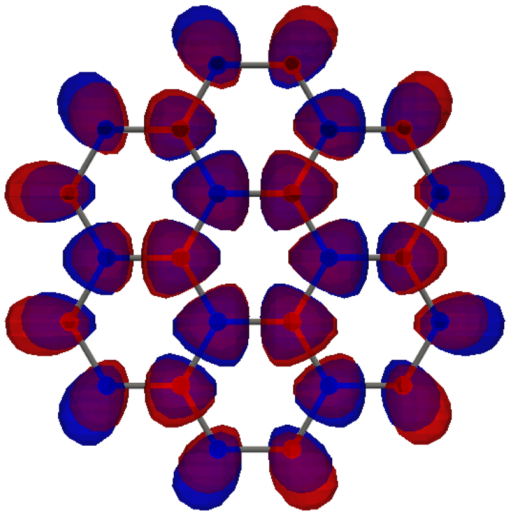
THE UNIVERSITY OF
MELBOURNE

Final Report

The Art of Scientific Computing: Finite Graphene Flakes using Hückel Theory

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Contents

1 Introduction **2**
- 1.1 Graphene and Motivation . . . 2
- 1.2 Finite Graphene Flakes and Edge Structure . . . 2
- 1.3 Hückel Molecular Orbital Theory . . . 3
- 1.4 Project Aims and Objectives . . . 4

2 Theory **5**
- 2.1 Structure of Graphene . . . 5
- 2.2 The Hückel Hamiltonian . . . 5
- 2.3 Molecular Orbital Occupation and Energy Gaps . . . 6
- 2.4 Finite-Size Effects and Sublattice Symmetry . . . 6
- 2.5 Molecular Orbital Reconstruction . . . 7

3 Computational Methodology **8**
- 3.1 Lattice Construction . . . 8
- 3.2 Electronic Structure Calculations . . . 8
- 3.3 Molecular Orbital Visualisation . . . 8

4 Results and Discussion **9**
- 4.1 Benzene: Verification and Orbital Structure . . . 9
- 4.2 Molecular Orbital Energy Spectra . . . 9
- 4.3 HOMO-LUMO Gap: Completed Shells . . . 11
- 4.4 Incomplete Shells and Zero-Energy States . . . 12
- 4.5 Coronene: Orbital Visualisation . . . 13

5 Model Limitations and Further Work **15**
- 5.1 Model Limitations . . . 15
- 5.2 Applications and Future Work . . . 15

6 Conclusion **16**

Bibliography **16**

Chapter 1

Introduction

1.1 Graphene and Motivation

Graphene is comprised of a single 2D layer of carbon atoms arranged in a honeycomb lattice. Since it's experimental isolation in 2004 (Novoselov et al. 2004), graphene has been studied extensively due to it's unique structure and properties. It's high mechanical strength, thermal conductivity, and unusual electronic behaviour makes it a promising material for a wide range of applications including high speed electronics, biomedicine, photonics and energy storage (Singh 2025). However, the practical use of graphene in electronic devices also depends on the ability to control its electronic energy levels.

In graphene, each carbon atom in graphene bonds to three of its neighbours through sp^2 hybridisation, leaving one unhybridised $2p_z$ orbital perpendicular to the lattice plane. The overlap between these orbitals produces a delocalised π-electron system responsible for much of graphene's electronic behaviour. In an ideal, infinite graphene sheet, the valence and conduction bands meet at the Dirac points, giving a vanishing band gap (Castro Neto et al. 2009). Although this electronic structure contributes to graphene's unusual transport properties, the absence of an intrinsic band gap makes conventional semiconductor-like switching difficult.

A proposed approach to controlling graphene's electronic structure is to restrict its spatial dimensions. Graphene nanoribbons, which are confined in one in plane direction, have been investigated as a means of introducing and controlling electronic energy gaps. Both theoretical calculations and experimental measurements have demonstrated the strong dependence of nanoribbon energy gaps on their width and edge structure (Son, Cohen, and Louie 2006; Han et al. 2007).

Further confinement produces graphene nanoflakes or graphene quantum dots, which are finite in both in plane directions. Unlike the continuous electronic bands of an infinite periodic system, finite structures possess discrete molecular energy levels. The separation between these levels depends on the number of carbon atoms, the geometry of the flake and its boundary conditions. The energy separation between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) provides a useful measure of the frontier level separation within a molecular orbital model (Jeong and Lee 2024).

For many families of similarly shaped graphene nanoflakes, the HOMO-LUMO gap decreases as the system becomes larger. The investigation of this size dependence provides a connection between the molecular orbital description of small aromatic compounds and the electronic structure of extended graphene (Jeong and Lee 2024).

1.2 Finite Graphene Flakes and Edge Structure

Finite graphene flakes provide a useful system for investigating how electronic structure changes as a conjugated carbon network increases in size. Small aromatic molecules, including benzene and coronene, can be regarded as finite fragments of the graphene honeycomb lattice. By progressively adding fused hexagonal rings, increasingly large structures can be generated while preserving the same underlying carbon connectivity.

In this project, graphene-like structures are constructed by adding successive shells of benzene rings around a central hexagon. The resulting completed shell structures form a family of hexagonal flakes for which the number of carbon sites is

$$N = 6m^2, \tag{1.1}$$

where m denotes the number of completed shells, including the central ring. The first three members of this sequence contain $N = 6$, 24 and 54 carbon atoms, corresponding to benzene, coronene and circumcoronene, respectively (Jeong and Lee 2024; Zou et al. 2023). This sequence provides a systematic means of comparing the electronic spectra for increasingly larger flakes while preserving their overall hexagonal geometry.
The electronic properties of finite graphene structures depend not only on their size, but also on the arrangement of carbon atoms along their boundaries. Two important edge configurations are known as zigzag and armchair edges. These possess different atomic connectivities and can consequently support different electronic states. In particular, zigzag edges can support electronic states concentrated near the boundary, with energies close to the Fermi level (Nakada et al. 1996). Armchair edges generally exhibit different electronic behaviour and have a dependence on ribbon width (Son, Cohen, and Louie 2006).

The completed shell structures considered in this project include hexagonal zigzag edge graphene flakes, as examined by Jeong and Lee (Jeong and Lee 2024). Circumcoronene, for example, possesses six zigzag terminated sides (Zou et al. 2023). These structures therefore provide a consistent group of flakes for examining how electronic energy levels evolve with increasing size while retaining the same broad lattice symmetry and edge type.

However, limiting the investigation to completed shells leaves a relatively small number of possible system sizes. To explore the electronic structure between these configurations, the computational model also permits incomplete outer shells. Additional hexagonal rings are introduced sequentially and, where necessary, only part of the final ring is incorporated to achieve a specified number of carbon atoms.

These intermediate structures provide an opportunity to investigate how changes in atomic connectivity, boundary geometry and the number of electronic states influence the calculated spectrum. They are not necessarily chemically stable molecules, and changes in size and geometry cannot be varied independently within this construction. Consequently, they are treated as exploratory mathematical models rather than direct predictions for physical nanoflakes.

An additional consideration is the occupation of molecular orbitals. Within the π-electron model, each carbon atom contributes one electron. Structures containing an odd number of carbon atoms therefore have an odd number of π electrons, requiring a treatment of singly occupied orbitals that is not included in the present closed shell analysis. Accordingly, the systematic HOMO-LUMO gap calculations are restricted to even numbers of carbon sites. Possible zero energy states and degeneracies in the resulting single particle spectra must also be considered when interpreting the calculated frontier level separations.

1.3 Hückel Molecular Orbital Theory

The electronic structure of graphene nanoflakes can be investigated using several computational approaches, ranging from first principles electronic structure methods to simplified tight binding models. Although more sophisticated methods can account for additional physical interactions, their computational cost can limit the number and size of structures that can be examined systematically.

In this project, the π-electron system is modelled using Hückel molecular orbital (HMO) theory, originally developed to describe conjugated molecules and presented in detail by Streitwieser (1964). In its simplest form, the method represents each carbon atom by one $2p_z$ orbital and includes interactions only between nearest neighbour carbon atoms. The resulting Hamiltonian is determined by the connectivity of the lattice, allowing electronic energy levels and molecular orbital coefficients to be calculated through numerical matrix diagonalisation.

This approach is particularly appropriate for the present investigation as the same Hamiltonian construction can be applied to both small aromatic molecules and larger graphene structures. Furthermore, the nearest neighbour Hückel approximation is mathematically equivalent to the simplest single orbital tight-binding model of graphene when corresponding parameters are chosen consistently (Castro Neto et al. 2009). It therefore provides a direct connection between molecular orbital theory and the electronic models commonly employed in condensed matter physics.

Previous calculations by Jeong and Lee (Jeong and Lee 2024) demonstrate that the Hückel method can reproduce important features of graphene nanoflake energy spectra, including changes in orbital energies and

HOMO–LUMO separations with increasing system size. Its computational simplicity also makes it useful for investigating the influence of different geometries without requiring a separate electronic structure calculation for every system.

Nevertheless, the approximation neglects explicit electron–electron interactions, spin polarisation and longer range hopping. Consequently, the calculated eigenvalues describe an approximate single particle electronic structure, and the HOMO-LUMO separation should not be identified directly with an experimental optical or quasiparticle gap. Within these limitations, the model provides a useful means of investigating the relationship between lattice geometry and the distribution of electronic energy levels.

The underlying theory, approximations and mathematical construction of the Hamiltonian are developed in Chapter **??**.

1.4 Project Aims and Objectives

The aim of this project is to determine how the electronic structure of finite hexagonal graphene flakes evolves with increasing system size, and to assess the extent to which the Hückel approximation reproduces the trends reported in the literature. Particular attention is given to the HOMO-LUMO gap, the distribution of molecular orbital energies and the influence of incomplete outer shells.

To investigate these questions, a Python program is developed to generate finite honeycomb lattices, identify nearest neighbour carbon atoms and construct the corresponding Hückel Hamiltonian. Numerical diagonalisation provides the molecular orbital energies and coefficients, which are subsequently used to calculate frontier level separations and visualise selected molecular orbitals.

The specific objectives of the project are:

1. To construct and verify a numerical Hückel model for finite graphene flake structures, using known results for small aromatic molecules and published graphene nanoflake calculations as benchmarks.
2. To investigate the evolution of the molecular orbital energy spectrum as the number of carbon atoms increases, including changes in level spacing and degeneracy.
3. To determine the dependence of the HOMO-LUMO gap on the number of carbon sites, comparing the completed shell series with intermediate, incomplete shell structures. The completed shell results are also examined for systematic size scaling behaviour.
4. To visualise selected molecular orbitals using 3D isosurfaces and 2D contour plots, and relate their spatial distributions to the underlying carbon lattice.
5. To compare the calculated results with the literature and critically evaluate the capabilities and limitations of the nearest neighbour Hückel approximation.

Through these investigations, the project aims to establish how much of the size dependent electronic behaviour of finite graphene flakes can be understood from the connectivity of their π-electron network alone. The results provide a basis for interpreting the transition from small aromatic molecules towards larger graphene flake structures, while identifying the physical effects that require more sophisticated theoretical treatments.

The following chapter introduces the theoretical framework of the Hückel approximation. Chapter **??** describes the computational implementation and structure-generation algorithm, while Chapter **??** presents the numerical results and comparisons with previous work.

Chapter 2

Theory

2.1 Structure of Graphene

Graphene is a single atomic layer of carbon atoms arranged in a honeycomb lattice. As the honeycomb structure is not a Bravais lattice type itself, it may be viewed as two interpenetrating triangular sublattices, conventionally labelled A and B (see Figure 2.1) (Castro Neto et al. 2009). Each unit cell therefore contains two carbon atoms, one belonging to each sublattice.

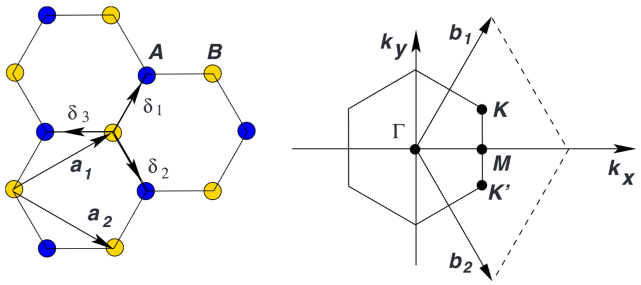


Figure 2.1: Honeycomb lattice of graphene, shown as two interpenetrating triangular sublattices A and B, from (Castro Neto et al. 2009).

This sublattice structure is central to the electronic properties of finite graphene flakes. In particular, changes in edge structure or in the relative numbers of A and B sites can affect the states near the Fermi level, including the appearance of zero-energy states in some finite graphene nanostructures (Fernández-Rossier and Palacios 2007; Potasz, Güçlü, and Hawrylak 2010).

2.2 The Hückel Hamiltonian

Within the Hückel approximation, the π-electron system of a graphene flake containing N carbon atoms is represented by N interacting $2p_z$ atomic orbitals. As described in the course notes, the overlap matrix is approximated by the identity, and interactions are restricted to nearest neighbour carbon atoms. The resulting Hamiltonian may be expressed in terms of the lattice adjacency matrix $\mathbf{A}$,

$$\mathbf{H} = \alpha\mathbf{I} + \beta\mathbf{A} \tag{2.1}$$

where $A_{ij} = 1$ if atoms i and j are bonded and zero otherwise. The parameter α defines the on-site orbital energy, while β describes the nearest neighbour interaction. The Hamiltonian therefore depends on the connectivity of the carbon lattice, allowing structures with different geometries to be investigated using the same model.

The molecular orbital energies E_k and coefficient vectors $\mathbf{c}_k$ are obtained by solving

$$\mathbf{H}\mathbf{c}_k = E_k\mathbf{c}_k. \tag{2.2}$$

Each eigenvalue corresponds to a molecular orbital energy, while the elements c_{ik} of its eigenvector describe the contribution of carbon site i to orbital k.

For the calculations presented here, $\alpha = 0$ and $\beta = -1$, giving orbital energies in units of $|\beta|$. For comparison with the literature, a hopping magnitude of $|\beta| = 2.40$ eV is adopted (Jeong and Lee 2024). This energy scale allows the dimensionless separations obtained from diagonalisation to be expressed in electronvolts without altering the underlying model.

2.3 Molecular Orbital Occupation and Energy Gaps

For a neutral graphene flake, each carbon atom contributes one π electron. Assuming an even number of electrons and a closed shell ground state, the lowest $N/2$ spatial molecular orbitals are doubly occupied. For eigenvalues ordered by increasing energy, the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) are therefore identified by

$$E_{\mathrm{HOMO}} = E_{N/2}, \qquad E_{\mathrm{LUMO}} = E_{N/2+1}. \tag{2.3}$$

The corresponding energy separation is

$$\Delta E_{\mathrm{HL}} = E_{\mathrm{LUMO}} - E_{\mathrm{HOMO}}. \tag{2.4}$$

The HOMO-LUMO gap provides a convenient measure of the separation between the frontier electronic states and allows the effects of flake size and geometry to be examined quantitatively.

For structures with an odd number of π electrons, a singly occupied orbital must be considered. Furthermore, even electron systems may exhibit degeneracies at the occupation boundary, in which case a simple closed shell description may not adequately represent the electronic ground state (Fernández-Rossier and Palacios 2007). The systematic gap analysis is therefore restricted to even numbers of carbon atoms, with zero or degenerate frontier level separations interpreted within the limitations of the single particle model.

Importantly, the calculated HOMO-LUMO gap is not generally equivalent to an experimental optical or quasi-particle gap, since electron-electron interactions and excited state effects are not explicitly included in the Hückel approximation.

2.4 Finite-Size Effects and Sublattice Symmetry

The discrete energy spectrum of a finite graphene flake depends on both the number of carbon atoms and their connectivity. As the system increases in size, the number of available molecular orbitals increases, and the energy levels generally become more closely spaced. However, the detailed behaviour of the frontier orbitals also depends on the geometry and symmetry of the flake.

For geometrically similar 2D structures, the number of carbon sites scales approximately with the flake area, $N \propto L^2$, where L is a characteristic linear dimension. Since the low energy dispersion of ideal graphene is approximately linear near its Dirac points, confinement motivates a characteristic energy scale proportional to $1/L$ (Castro Neto et al. 2009). This suggests a possible relationship

$$\Delta E_{\mathrm{HL}} \propto \frac{1}{L} \propto N^{-1/2}. \tag{2.5}$$

This scaling is not universal, as the frontier level separation can be strongly influenced by edge states and molecular symmetry. Nevertheless, it provides a physically motivated relationship against which the completed shell results can be compared.

A further property of the graphene honeycomb lattice is its bipartite structure. Carbon sites can be divided into two sublattices, A and B, such that nearest neighbour interactions occur only between opposite sublattices. Consequently, the nearest neighbour Hamiltonian has the block structure

$$\mathbf{H} - \alpha\mathbf{I} = \begin{pmatrix} \mathbf{0} & \beta\mathbf{B} \\ \beta\mathbf{B}^T & \mathbf{0} \end{pmatrix}, \tag{2.6}$$

where $\mathbf{B}$ describes the connections between the two sublattices.

This symmetry requires the energy spectrum to occur in pairs about α. When the sublattices contain unequal numbers of atoms, at least $|N_A - N_B|$ states occur at the on site reference energy. Additional zero energy states may also arise from the lattice connectivity, even when the sublattices are balanced.

These properties are relevant to the intermediate structures considered in this project. Incomplete outer rings can alter the sublattice populations, local coordination and overall symmetry, potentially introducing zero energy states or changing the frontier level separation (Fernández-Rossier and Palacios 2007). Consequently, deviations from the completed shell trend cannot necessarily be attributed to size alone.

The importance of boundary geometry is also evident in studies of zigzag edged graphene structures, where states localised near the edges can occur close to the Fermi energy (Nakada et al. 1996). These effects motivate examining both the energy eigenvalues and spatial character of the calculated molecular orbitals.

2.5 Molecular Orbital Reconstruction

While diagonalisation of the Hückel Hamiltonian gives molecular orbital energies and coefficients, the eigenvectors describe amplitudes only at discrete carbon sites. To visualise the orbitals in 3D space, these coefficients are combined with approximate atomic $2p_z$ functions through the linear combination of atomic orbitals (LCAO),

$$\psi_k(\mathbf{r}) = \sum_{i=1}^{N} c_{ik}\phi_i(\mathbf{r}), \tag{2.7}$$

where $\phi_i(\mathbf{r})$ is a Slater-type orbital centred on a carbon atom i, as defined in the course notes. In the present model, these functions have p_z symmetry, with angular momentum quantum numbers $l = 1$ and $m = 0$.

The reconstructed wavefunctions can be represented by 3D isosurfaces or 2D contour plots. Positive and negative wavefunction values correspond to opposite orbital phases, rather than positive and negative electronic charges.

The relative contribution of each carbon site to the molecular orbital k is described by $|c_{ik}|^2$, providing a means of examining whether an orbital is distributed throughout the flake or concentrated near its boundaries. However, for degenerate orbital energies, the individual eigenvectors are not unique and their spatial appearance may depend on the particular basis chosen within the degenerate subspace.

Chapter 3

Computational Methodology

The Hückel model was implemented in Python using NumPy and SciPy for numerical calculations, Matplotlib for 2D plotting and PyVista for 3D molecular orbital visualisation. The program was organised into reusable functions for lattice generation, Hamiltonian construction, eigenvalue analysis and graphical output.

3.1 Lattice Construction

Finite graphene structures were generated by constructing a central benzene ring and adding successive shells of neighbouring hexagons. Carbon positions were calculated using a fixed nearest neighbour bond length of $a_{\mathrm{CC}} = 1.42$ Å, with duplicate atomic positions removed to preserve the honeycomb lattice. Completed shells contain $N = 6m^2$ carbon atoms, where m denotes the number of completed shells. Intermediate structures were generated by partially filling the next outer ring until the requested site count was reached. These geometries were not subjected to structural optimisation and therefore represent idealised carbon networks.

Nearest neighbour bonds were identified from the atomic coordinates using a spatial search with a bond length tolerance of 0.10 Å. The resulting connectivity was used to construct the symmetric Hückel Hamiltonian, with diagonal elements $\alpha = 0$ and nonzero off diagonal elements $\beta = -1$ for bonded carbon pairs.

3.2 Electronic Structure Calculations

The Hamiltonian was diagonalised numerically to obtain the molecular orbital energies and eigenvectors. For even N, the HOMO and LUMO were identified from the two central eigenvalues, and their separation was converted to electronvolts using $|\beta| = 2.40$ eV. For calculations requiring only the energy gap, SciPy was used to evaluate these two eigenvalues without computing the complete set of eigenvectors, reducing computational expense. The underlying methodology and original implementation were developed by the author, while ChatGPT was consulted to identify and implement computational optimisations.

The HOMO-LUMO gap was evaluated for complete shell structures up to $N = 294$ and for intermediate even site counts over the same range. Calculated gaps were reused across the comparison plots to avoid redundant diagonalisation. The complete shell results were additionally fitted to a power law, $\Delta E_{\mathrm{HL}} \propto N^{-p}$, using linear regression in logarithmic space, allowing comparison with the theoretical $N^{-1/2}$ scaling estimate.

3.3 Molecular Orbital Visualisation

The calculated eigenvectors were combined with Slater-type $2p_z$ atomic functions to reconstruct molecular orbitals on a 3D Cartesian grid. Positive and negative isosurfaces were extracted at 20% of the maximum absolute wavefunction amplitude for each orbital and visualised using PyVista, with carbon atoms and nearest neighbour bonds superimposed. Additional orthogonal views and two-dimensional contour plots were generated to examine orbital phase and nodal structure.

Orbital reconstruction was performed separately from the Hamiltonian diagonalisation and did not affect the calculated eigenvalues. Selected visualisations of benzene and coronene were used to illustrate the spatial characteristics of bonding, antibonding and frontier molecular orbitals.

Chapter 4

Results and Discussion

The electronic structure of finite graphene flakes was investigated using the nearest neighbour Hückel model, with $\alpha = 0$ and $\beta = -1$. The calculated energy separations were converted into electronvolts using $|\beta| = 2.40$ eV, following Jeong and Lee (2024). Complete shell structures containing $N = 6m^2$ carbon sites were examined to establish size dependent trends and for quantitative comparisons with literature. Intermediate even site counts were also considered to investigate the influence of incomplete outer shells on the electronic spectrum.

4.1 Benzene: Verification and Orbital Structure

Benzene ($N = 6$) provides an analytical benchmark for validating the numerical implementation and examining the spatial structure of the molecular orbitals. The calculated eigenvalues are

$$\frac{E_k}{|\beta|} = \{-2, -1, -1, 1, 1, 2\}, \tag{4.1}$$

in agreement with the analytical Hückel solution given in the course notes. The three lowest spatial orbitals are occupied by six π electrons, giving a HOMO-LUMO separation of $2|\beta|$, or 4.80 eV. The agreement in energies and degeneracies confirm that the Hamiltonian construction and diagonalisation reproduce the expected result.

The corresponding orbital visualisations are shown in Figure 4.1. The lowest energy orbital (1) has coefficients of equal magnitude and phase in all six carbon atoms, representing a fully bonding combination of the $2p_z$ orbitals. In contrast, the highest-energy orbital (6), exhibits alternating phases around the ring, corresponding to an antibonding arrangement with destructive overlap between neighbouring atomic orbitals.

The two degenerate orbitals at $E = -|\beta|$ constitute the HOMO level, while those at $E = +|\beta|$ constitute the LUMO level. Individual eigenvectors within these degenerate subspaces are not unique, so their spatial representations depend on the orthonormal basis selected. The reconstructed orbitals also display the characteristic $2p_z$ nodal plane at $z = 0$, with positive and negative wavefunction phases extending above and below the carbon framework. These phases describe the sign of the wavefunction rather than electric charge.
Benzene therefore provides both a numerical validation of the model and a reference for interpreting the more complex orbital structures of larger graphene flakes.

4.2 Molecular Orbital Energy Spectra

Figure 4.2 compares the calculated molecular orbital spectra of completed shell flakes containing 6, 24, 54 and 96 carbon atoms. As the flake size increases, the number of available molecular orbitals increases and so the energy levels become more densely distributed. Increasing the number of states near the occupation boundary reduces the HOMO-LUMO separation.

The spectra are symmetric about $E = 0$, as expected for the bipartite nearest neighbour Hamiltonian with $\alpha = 0$. Degenerate eigenvalues are also present, reflecting the symmetry of the completed shell geometries.

These observations agree qualitatively with the Hückel calculations of Jeong and Lee (2024), who identify increasingly dense molecular energy spectra and degeneracies around $E = \pm|\beta|$ for larger graphene nanoflakes. Similar discrete electronic shell structures have been investigated using tight-binding calculations by Heiskanen,

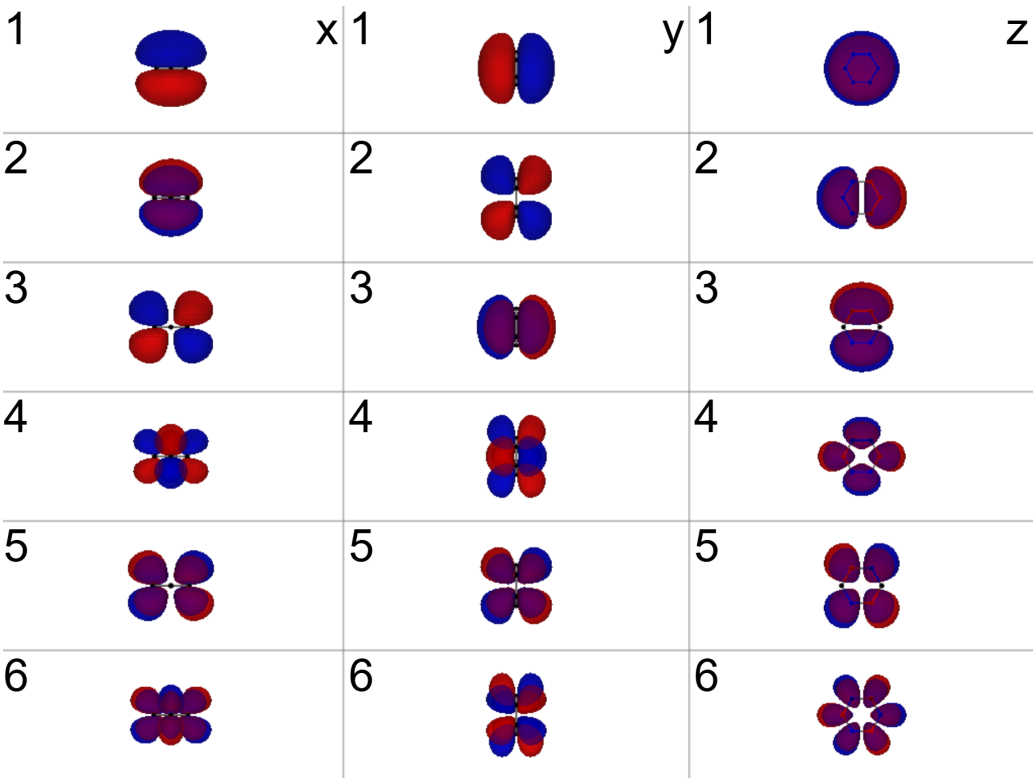


Figure 4.1: Hückel molecular orbitals for benzene, viewed from the yz, zx and xy planes. Red and blue indicate opposite wavefunction phase. Carbon atoms and bonds indictate the underlying moelcular geometry.

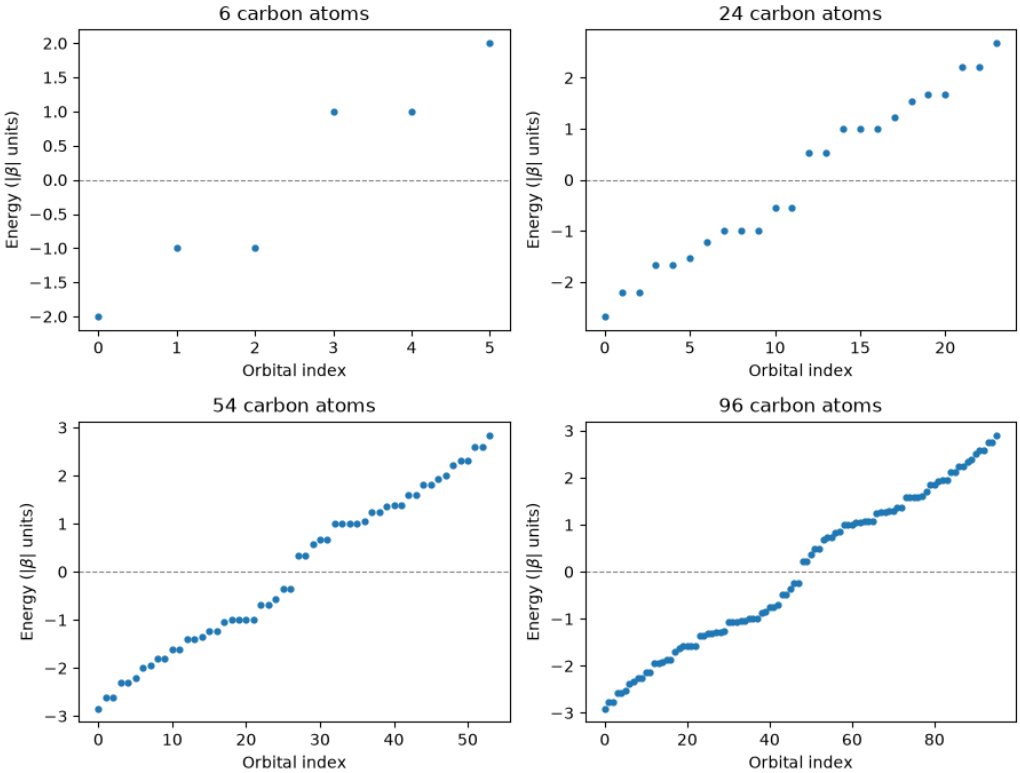


Figure 4.2: Calculated Hückel molecular orbital energy spectra for $N = 6$, 24, 54 and 96 carbon atoms. Energies are expressed in units of $|\beta|$, relative to $\alpha = 0$.

Manninen, and Akola (2008).

Importantly, these spectra represent the ordered eigenvalues of finite Hamiltonian matrices. They should not

be interpreted as electronic band structures, since the horizontal axis is the molecular orbital index rather than a crystal momentum.

4.3 HOMO-LUMO Gap: Completed Shells

The HOMO-LUMO gap was calculated for the first seven completed shell structures, containing between 6 and 294 carbon atoms. The results are presented in Table 4.1 and Figure 4.3.

Table 4.1: Calculated HOMO-LUMO separations for completed shell graphene flakes, using $|\beta| = 2.40$ eV. Published reference values for the corresponding Hückel systems are included where available.

N	$\Delta E_{\mathrm{HL}}/\lvert\beta\rvert$	ΔE_{HL} (eV)	Literature (eV)
6	2.000000	4.800	4.80
24	1.078378	2.588	–
54	0.684082	1.642	1.64
96	0.458897	1.101	1.10
150	0.313382	0.752	–
216	0.214153	0.514	–
294	0.145213	0.349	–

Literature values are from Jeong and Lee (2024).

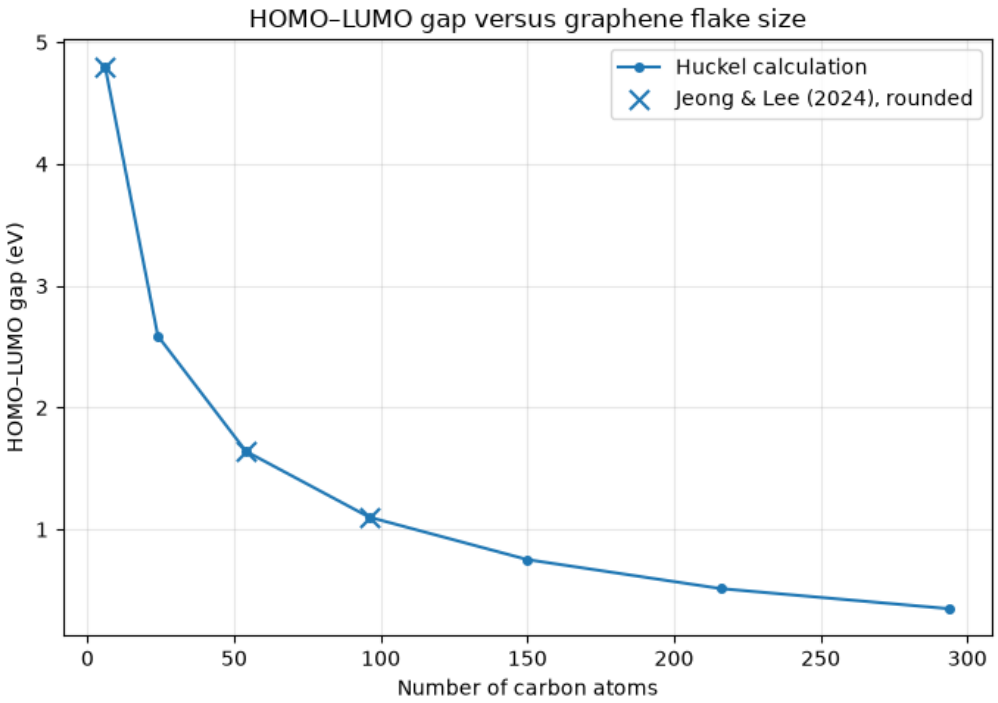


Figure 4.3: HOMO-LUMO gap as a function of carbon site count for completed shell graphene flakes. The present numerical results are compared with the rounded Hückel values reported by Jeong and Lee (2024).

As the lattice size increases, the HOMO-LUMO gap decreases. The gap decreases from 4.800 eV for benzene to 0.349 eV for the 294-carbon flake, representing a reduction of approximately 92.7%. Although the gap becomes substantially smaller, it remains finite for the largest structure considered. This behaviour is consistent with the approach towards the gapless electronic spectrum of ideal, infinite graphene, while retaining the effects of the finite size and boundary geometry.

The calculated results closely agree with published Hückel values for the corresponding hexagonal zigzag edged nanoflakes. Jeong and Lee (2024) report gaps of approximately 1.64 eV for the 54-carbon structure and 1.10 eV for the 96-carbon structure. The present calculations give 1.6418 and 1.1014 eV, respectively, differing from these rounded published values by less than 0.002 eV.

For coronene, the calculated gap of $1.07838|\beta|$ agrees with the non-interacting Hückel result of $1.07838t_0$ reported by Prodhan, Mazumdar, and Ramasesha (2019), where t_0 denotes the hopping magnitude.
The close numerical agreement provides evidence that the present implementation correctly reproduces the electronic structure of these systems within the specified Hückel approximation. However, agreement between calculations based on essentially the same Hamiltonian does not establish equivalent agreement with experimental gaps.

4.3.1 Gap Scaling

The observed reduction in the HOMO-LUMO gap is qualitatively consistent with the finite size scaling argument discussed in Section 2.4. To investiagte this, the completed shell gaps were fitted to a power law of the form

$$\Delta E_{\mathrm{HL}} \propto N^{-p}. \tag{4.2}$$

A linear fit to $\log(\Delta E_{\mathrm{HL}})$ against $\log N$ for the seven completed shell structures gives $p \approx 0.660$, with $R^2 \approx 0.965$, as shown in Figure 4.4.

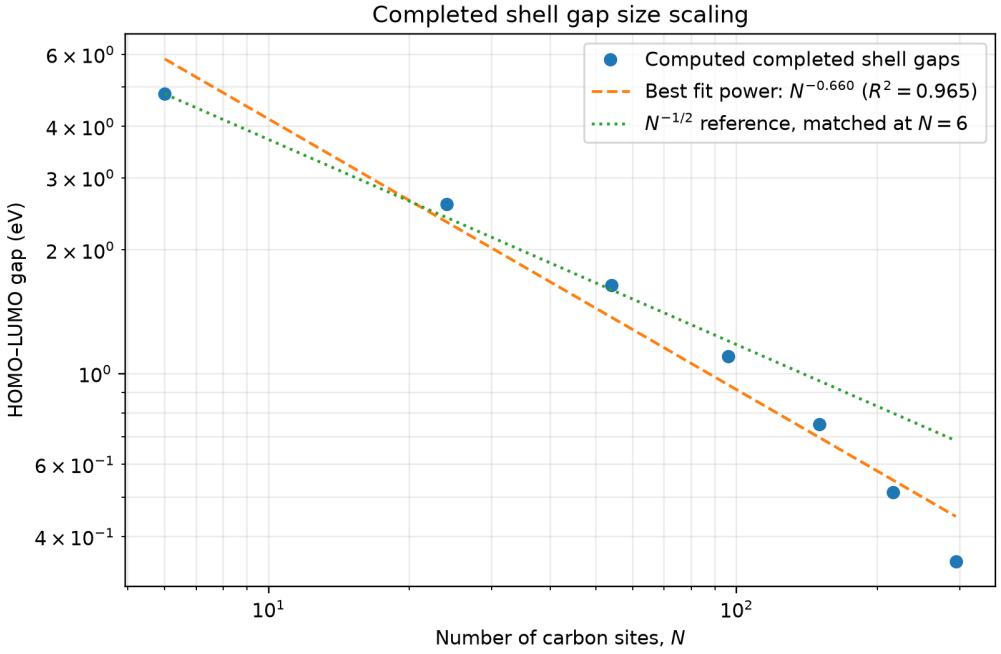


Figure 4.4: Log-log plot of the HOMO-LUMO gap against the number of carbon atoms for seven completed shell graphene flakes ($N = 6$–294). The calculated values are compared with a fitted power law, $\Delta E_{\mathrm{HL}} \propto N^{-0.660}$ ($R^2 = 0.965$), and the theoretical $N^{-1/2}$ scaling reference, normalised to the benzene gap.

The fitted exponent is larger than the approximate $p = 1/2$ value suggested by simple 2D confinement arguments. The discrepancy may reflect the importance of finite size corrections, boundary geometry and the detailed distribution of electronic states. The fitted exponent is also sensitive to the restricted range of flake sizes considered, and should therefore not be interpreted as a universal scaling exponent for graphene.

The analysis demonstrates that the gap decreases systematically for this group of hexagonal flakes, providing a quantitative extension of the trends reported by Jeong and Lee (2024).

4.4 Incomplete Shells and Zero-Energy States

While the completed shell can be used to investigate size dependence, it does not reveal how the electronic spectrum changes as individual carbon sites are added between successive completed configurations. To explore this behaviour, the HOMO-LUMO separation was also calculated for every even value of N between 6 and 294.

Figure 4.5 shows that the intermediate structures exhibit greater variation than the completed shell sequence. Although the general trend towards smaller gaps remains evident, the frontier level separation is not monotonic with increasing carbon site count.

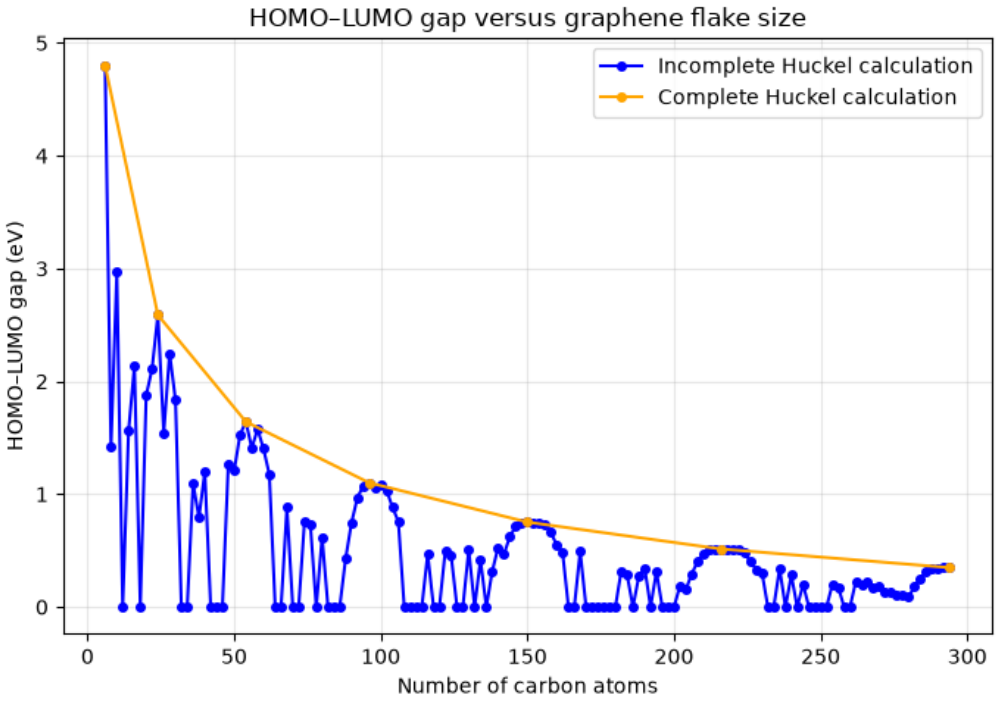


Figure 4.5: Central eigenvalue separation for even carbon site counts from $N = 6$ to 294, with completed shell configurations highlighted.

For example, as the calculated gap decreases to 1.544 eV at $N = 26$, yet increases to 2.249 eV at $N = 28$, before decreasing to 1.835 eV at $N = 30$. At $N = 32$, the separation between the central eigenvalues vanishes. Thus, the addition of carbon atoms does not necessarily reduce the gap using this model.

The zero gap obtained at $N = 32$ can be understood through the bipartite structure of the generated carbon network. The two sublattices contain $N_A = 17$ and $N_B = 15$ sites, giving an imbalance of $|N_A - N_B| = 2$. This guarantees at least two zero energy eigenstates in the nearest neighbour model. The calculated spectrum contains exactly two such states, resulting in a vanishing separation between the central eigenvalues. This behaviour agrees with the relationship between sublattice imbalance and zero-energy states discussed by Fernández-Rossier and Palacios (2007).

These results demonstrate that the electronic spectrum depends on the connectivity and symmetry of the carbon network, rather than site count alone. Incomplete shell growth changes the number of atoms together with their local coordination, boundary configuration and sublattice populations. Consequently, the fluctuations cannot be attributed solely to changes in flake size.

Furthermore, the intermediate structures are computationally generated carbon networks and need not correspond to stable, experimentally realisable molecules. A zero separation in the non-interacting Hamiltonian therefore does not establish metallic behaviour, since electron-electron interactions and spin-dependent effects may alter the electronic ground state.

4.5 Coronene: Orbital Visualisation

The molecular orbital visualisations provide a spatial interpretation of the Hamiltonian eigenvectors, complementing the energy spectra and HOMO-LUMO analysis. Selected orbitals were reconstructed using Slater-type $2p_z$ functions and visualised as 3D isosurfaces, with the underlying carbon lattice superimposed.

For coronene ($N = 24$), the HOMO and LUMO correspond to molecular orbitals 12 and 13, respectively, using the one-based indexing convention adopted in the program. Their eigenvalues are approximately $-0.5392|\beta|$ and $+0.5392|\beta|$, giving the previously calculated gap of $1.0784|\beta|$. The molecular orbitals 1, 12, 13 and 24 for N=24 are shown in Figure 4.6

These frontier levels are twofold degenerate: molecular orbitals 11 and 12 share the HOMO energy, while 13

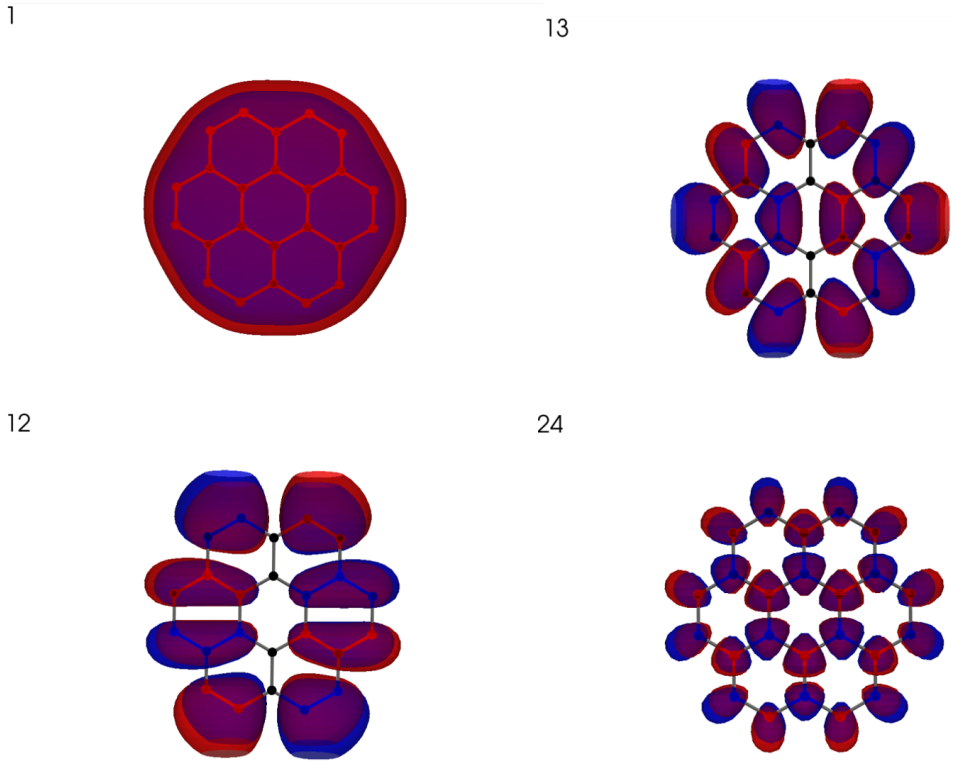


Figure 4.6: Selected Hückel molecular orbitals of coronene ($N = 24$). Molecular orbital (1) represents the lowest energy bonding orbital, (12) corresponds to a HOMO, (23) corresponds to a LUMO and (24) is the highest energy antibonding orbital. The underlying carbon atoms and bonds are superimposed.

and 14 share the LUMO energy. As discussed for benzene, the appearance of an individual orbital within a degenerate subspace depends on the chosen eigenvector basis. The spatial plots therefore illustrate valid orbital representations without uniquely defining the shape of each degenerate state.

Molecular orbital 24, the highest-energy π orbital of coronene, was selected for the report cover. Unlike the frontier orbitals, it represents a strongly antibonding combination at the upper end of the calculated spectrum, with an energy of approximately $+2.6751|\beta|$. Its visualisation illustrates the spatial and phase structure obtainable from the reconstructed molecular orbitals, although it is not directly involved in defining the HOMO-LUMO gap.

The orbital visualisations therefore complement the numerical eigenvalues by illustrating the different bonding and antibonding combinations present in finite conjugated carbon networks.

Chapter 5

Model Limitations and Further Work

5.1 Model Limitations

The results demonstrate that the nearest neighbour Hückel model reproduces several significant features of the finite electronic structure of graphene. In particular, the model reproduces published energy gaps for the selected hexagonal nanoflakes, predicts a systematic narrowing of the frontier level separation across the completed shell series, and reveals substantial sensitivity to lattice connectivity in intermediate configurations.

The agreement with previously published Hückel calculations primarily validates the numerical implementation. The model neglects explicit electron-electron interactions, structural relaxation, chemical edge termination and longer range hopping. These limitations are particularly relevant for irregular structures or systems with degenerate frontier orbitals, where electron correlation and spin dependent effects may influence the ground-state electronic configuration (Yeh et al. 2018).

These limitations are particularly significant for the larger or irregular structures, where frontier orbital degeneracy and possible open shell character may require a many electron treatment. Nevertheless, the model provides a framework for identifying systematic trends and selecting finite graphene structures for more detailed electronic structure calculations.

5.2 Applications and Future Work

Understanding how graphene flake size and geometry influence electronic structure is important for the development of graphene based nanomaterials, particularly in molecular electronics, optoelectronics and quantum dot applications (Son, Cohen, and Louie 2006; Castro Neto et al. 2009). Although the Hückel approximation cannot directly predict experimental device performance, its computational efficiency makes it useful for identifying electronic trends and screening candidate structures before applying more sophisticated methods. In particular, the non-monotonic HOMO-LUMO gaps observed for incomplete shells demonstrate the importance of atomic connectivity alongside system size. Future work could investigate alternative flake geometries and edge terminations, incorporate longer range hopping and electron-electron interactions, and account for structural relaxation. Comparison with density functional theory and experimental measurements would further establish the accuracy and applicability of the model.

Chapter 6

Conclusion

This project developed a computational implementation of Hückel molecular orbital theory to investigate the electronic structure of finite graphene flakes. The model successfully reproduced the analytical energy spectrum of benzene and demonstrated close agreement with published Hückel calculations for larger graphene structures. For completed shell flakes, the HOMO-LUMO gap decreased systematically from 4.80 eV for benzene ($N = 6$) to approximately 0.35 eV for $N = 294$, a reduction of 93%. A power-law fit yielded $\Delta E_{\mathrm{HL}} \propto N^{-0.660}$, indicating a strong size dependence, although the fitted exponent differs from the idealised $N^{-1/2}$ scaling estimate.

The investigation of incomplete outer shells revealed a more complex, non-monotonic dependence on carbon site count, demonstrating that electronic properties are influenced by lattice connectivity and symmetry rather than size alone. Notably, the $N = 32$ structure exhibited two zero-energy states associated with a sublattice imbalance of $|N_A - N_B| = 2$. Furthermore, the reconstruction of molecular orbitals using Slater-type functions provided a qualitative visualisation of their spatial distributions, phases and bonding characteristics, complementing the numerical energy analysis.

Despite its simplicity, the nearest neighbour Hückel approximation provides a useful framework for understanding electronic trends in finite conjugated carbon systems. Its limitations, particularly the neglect of electron-electron interactions, structural relaxation and realistic edge termination, restrict direct comparisons with experimental electronic properties. Future extensions could incorporate these effects or compare the present results with higher level electronic structure calculations. Nevertheless, the project demonstrates how a relatively simple computational model can reveal the relationship between atomic structure and electronic behaviour in finite graphene flakes.

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